

Grace — region audit of the Derived mass/ratio constants (2026-08-06, K1245 sweep)

Casey’s standing rule (K1245): σ/dev comparisons valid only WITHIN one region (interior $\leftrightarrow$ interior, boundary $\leftrightarrow$ boundary, exterior $\leftrightarrow$ exterior); otherwise trust the interior/discrete value. Demote a tier ONLY for a reason internal to the region (a named input), never a cross-region σ . Guard (both directions): “trust interior” must NOT become a license to re-inflate fitted/input-dependent values.

Scope reconciliation

Broad net found 37 Derived-tier entries with nonzero exterior dev%. Removing 2 non-particle (const_104 Poisson ratio, const_103 P/S wave-velocity — solid-mechanics T1314, a different region-logic) and ~4 duplicate derivations ($m_\pi \times 2$, $m_b \times 2$, $m_d \times 2$, $m_c \times 2$) $\rightarrow$ ≈ 31 particle mass/ratio constants, matching Keeper’s seed.

The finding: the rule CONFIRMS the corpus; it does not demote a wall

I read the input-suspect bin per-entry (NOT keyword — the keyword pass both false-flagged the muon and missed the real rider, so it can’t be trusted either direction):

entry	computation	verdict
Quark cascade const_106–109 (m_u, m_d, m_s, m_c)	forced BST-integer ratios chained to m_e (itself Derived via gravity anchor); e.g. $m_u = 3\sqrt{2} \cdot m_e$, $m_d = (13/6)m_u$	KEEP Derived (forced interior); dev% vs measured MS-bar = cross-region confirmation
Wolfenstein const_087/089/090 (A, $\bar{\rho}$, $\bar{\eta}$)	pure integer ratios: $A=4/5$, $\bar{\eta}=1/(2\sqrt{\text{rank}})$, $\bar{\rho}=1/(2\sqrt{(2n_C)})$ — no input	KEEP Derived; dev% = confirmation (incl. A at 3.1% — large confirmation, forced value)
EW scales const_007/012/013 (v , m_W , m_Z)	$v=36\pi^{10}m_e/7$, $m_W=n_C \cdot m_p/(8\alpha)$, $m_Z=m_W/\cos\theta_W$ ($\sin^2\theta_W=3/13$) — built from forced quantities $\times$ Derived m_e unit	KEEP Derived; dev% = confirmation
m_b const_110 ($m_b=(7/3) \cdot m_\tau$)	RATIO 7/3 forced; ABSOLUTE plugs in measured $m_\tau=1776.86$	ALREADY correctly split — ratio Derived / absolute Indicative, ratified 2026-08-03 (Grace). No action.

Self-catch (both directions)

I initially flagged const_110 as a new named-input demotion to route to Casey. Reading its register_status showed I had already split it on Aug 3 (ratio-Derived/absolute-Indicative, Casey-ratified). Caught before routing an already-resolved item. The corpus already applies Casey’s region-logic where an input genuinely rides — const_110 is the proof-of-pattern, not a miss.

Result

- No new demotions. The one genuine input-rider (const_110) was already split honestly.
- No re-inflation. Nothing fitted/input-dependent is pushed back up; const_110’s absolute value stays Indicative.
- Mechanical cleanup remaining (labeling, my lane, low-risk): on the ~30 forced-interior keeps, relabel the exterior dev% explicitly as “confirmation (🔒 tier, cross-region)” so no future reader mistakes a cross-region dev% for a tier signal. Batch pass — stage for EOD, not a governance item.

- Koide (const_159): unchanged and now *explained* — a ratio-of-ratios, projection-invariant (the running cancels), so it is directly exterior-comparable (Tier-1 exact), the one lepton relation that legitimately crosses regions.

Governance: nothing needs Casey — the sweep confirms tiers and cleans labeling, exactly the predicted honest-floor outcome. Nothing pushed.